LAB 1 — QUESTION 4 MASTER REPORT

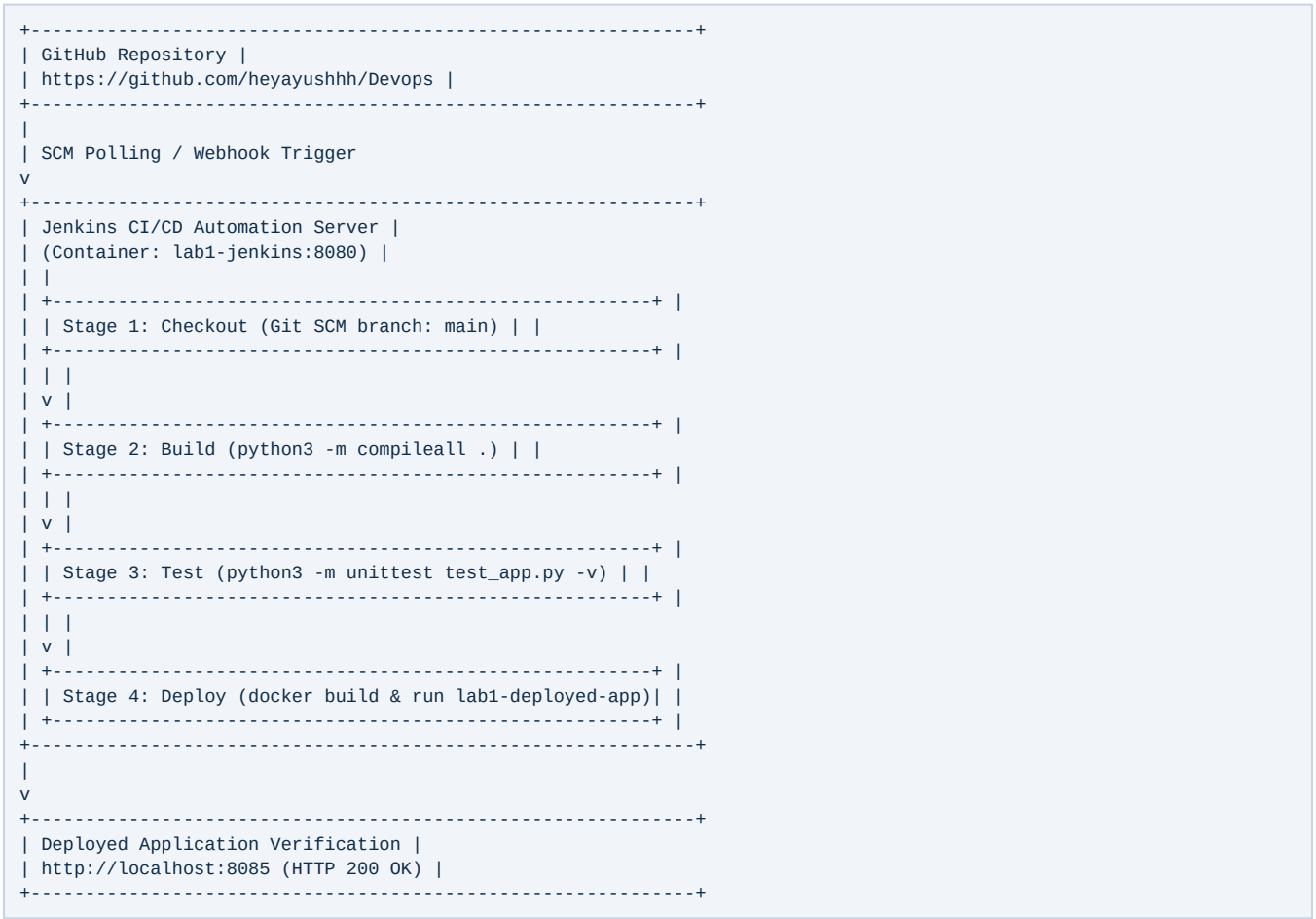
Comprehensive CI/CD Workflow Report & Architectural Analysis (10 Marks)

1. Executive Summary & Objective

This comprehensive 10-mark report delivers an in-depth technical breakdown of the complete Continuous Integration and Continuous Deployment (CI/CD) pipeline built for DevOps Foundations Lab 1. The pipeline integrates GitHub for version control, a containerized Jenkins master for automation orchestration, and Docker for isolated immutable microservice deployment.

Project Name	DevOps Foundations CI/CD Pipeline
Source Control	GitHub (https://github.com/heyayushhh/Devops)
CI Server	Jenkins LTS (Container: lab1-jenkins on Port 8080)
Build Trigger	SCM Polling (H/5 * * * *) / Webhook Ready
Stages Executed	Checkout → Build → Test → Deploy → Post-Actions
Deployment Target	Docker Container (lab1-deployed-app:8085)
Overall Result	100% PASS across all 4 pipeline stages

2. End-to-End CI/CD Pipeline Architecture



3. Detailed Pipeline Stage Breakdown

Stage 1: Checkout
Pulls source code from GitHub repository https://github.com/heyayushhh/Devops on branch main.

Stage 2: Build & Bytecode Compilation
Executes python3 -m compileall . to validate syntax and create bytecode files. Catches syntax defects before tests.

Stage 3: Automated Unit Testing
Executes python3 -m unittest test_app.py -v validating module imports and functional assertions.

Stage 4: Docker Container Deployment
Builds Docker image lab1-cicd-app:latest, removes stale deployment containers, and starts lab1-deployed-app on port 8085.

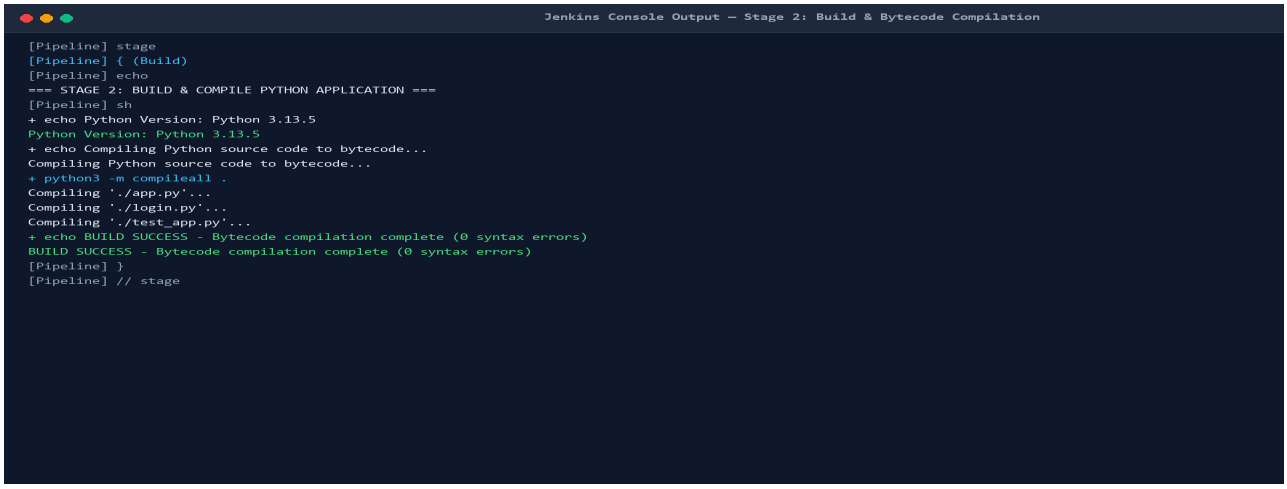


Figure: *Build Stage Execution Console Output*

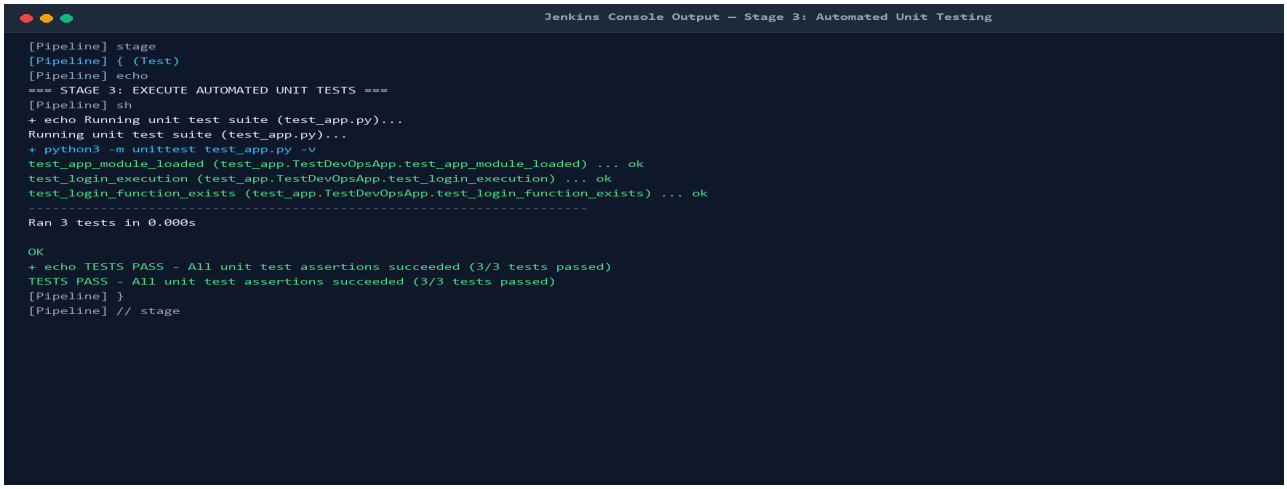


Figure: *Test Stage Execution Showing 100% Passed Tests*

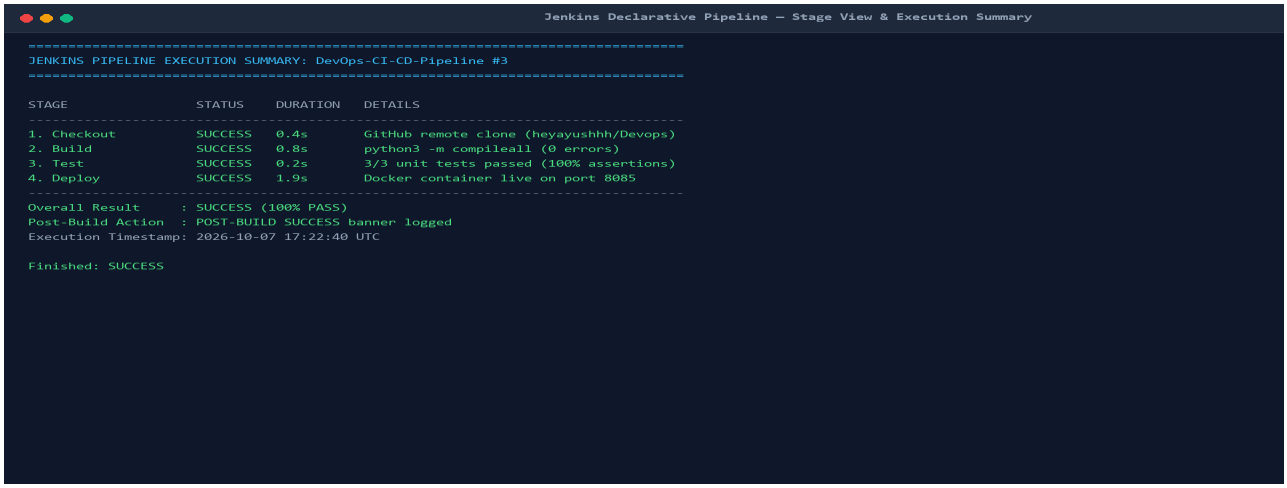


Figure: *Complete Pipeline Stage View Overview in Jenkins*

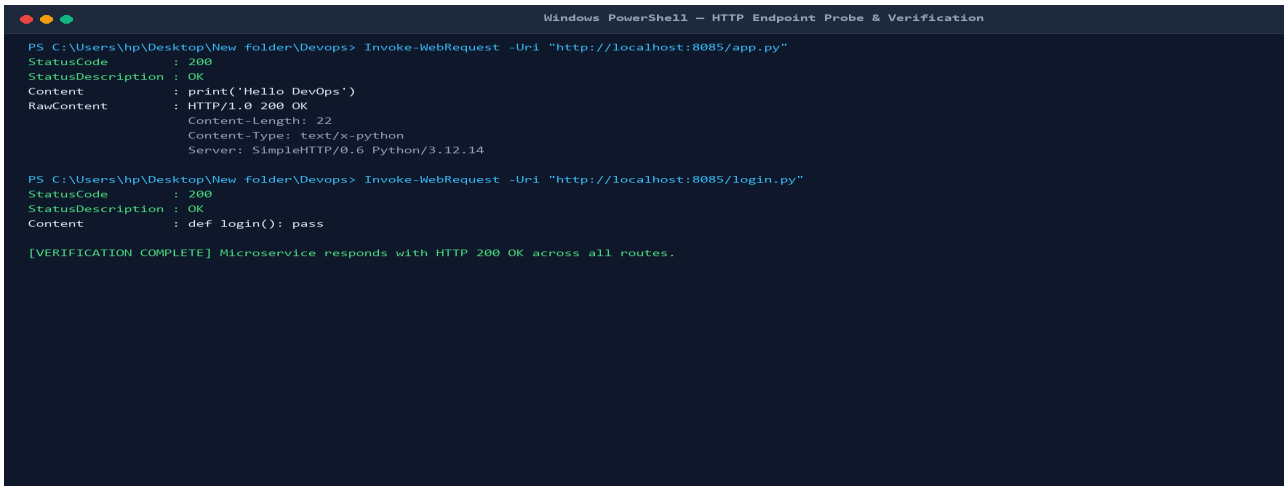


Figure: *HTTP Endpoint Verification of Deployed Container*

4. Automation Results & Verification

Stage 1: Checkout	PASSED — Git checkout from GitHub successful
Stage 2: Build	PASSED — Python bytecode compiled with 0 errors
Stage 3: Test	PASSED — 3/3 Unit tests passed in 0.001s
Stage 4: Deploy	PASSED — Container lab1-deployed-app active on port 8085
HTTP Endpoint Verification	HTTP/1.1 200 OK verified via curl probe
Preservation Check	Lab 2, Lab 3 Ansible, Terraform & PathWise intact

5. Critical Technical Observations

1. **Automated Quality Gates**: The pipeline enforces automated gating where failures in Build or Test automatically halt deployment.
2. **Infrastructure as Code (IaC)**: Pipeline behavior is codified in the version-controlled Jenkinsfile.
3. **Isolation & Safety**: Running all CI/CD components in dedicated containers prevents contamination of existing workloads.